Re-implementing "Distilling the Knowledge in a Neural Network" at Reduced Scale

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ABSTRACT

***Abstract*—Training large neural networks or ensembles of models improves generalization performance, but their high parameter count and computational latency hinder real-time deployment. Hinton et al. [1] address this limitation via "knowledge distillation," a compression framework that transfers the generalization capability of a cumbersome teacher model into a compact student network by training the student to match temperature-scaled "soft target" probability distributions. We re-implement temperature-scaled knowledge distillation from scratch in PyTorch on a single NVIDIA Tesla T4 GPU, transferring knowledge from an adapted ResNet-18 teacher ($11,172,810$ parameters) into a lightweight custom convolutional student ($38,282$ parameters, a $291.9\times$ compression ratio) on the MNIST dataset. Across a 5-epoch reduced-scale budget, the ResNet-18 teacher achieves 99.23% test accuracy (77 errors), the standalone hard-label student reaches 98.12% (188 errors), and the distilled student trained at $T = 10.0, \alpha = 0.1$ reaches 97.26% (274 errors). In a 4-epoch hyperparameter ablation across $T \in \{1, 2, 5, 10, 20\}$ and $\alpha \in \{0.0, 0.1, 0.5, 0.7, 0.9\}$, lower temperatures ($T = 1.0$, 97.81%) and higher hard-label weights ($\alpha = 0.9$, 98.04%) yield superior accuracy because the $291.9\times$ capacity-constrained student benefits from ignoring noisy negative logits at high temperatures, confirming the theoretical prediction in Section 2.1 of [1]. Most notably, we reproduce the held-out "mythical digit 3" experiment of Section 3 [1]: when all $6,131$ training examples of digit 3 are omitted from the transfer set, a hard-target student achieves 0.00% on test 3s ($1,010/1,010$ errors), whereas a distilled student trained solely on soft targets ($\alpha = 0.0, T = 10.0$) recovers 84.06% accuracy on unseen digit 3 ($161/1,010$ errors) and 95.55% overall test accuracy (445 errors) after post-training logit bias calibration ($+6.0$).**

***Index Terms*—knowledge distillation, model compression, temperature scaling, dark knowledge, soft targets, ResNet-18, reproducibility**

I. INTRODUCTION

In large-scale machine learning, there is a fundamental mismatch between the requirements of the training stage and the deployment stage [1]. Training aims to extract rich statistical structure from massive, redundant datasets where real-time latency is not required, favoring ensembles of multiple neural networks [3] or very large models regularized with dropout [5], [9]. Deployment to millions of end users, however, imposes strict constraints on memory footprint and inference latency. Buciluǎ, Caruana, and Niculescu-Mizil [1], [2] pioneered model compression by training a single small network to match the raw pre-softmax logits of a large ensemble. Hinton, Vinyals, and Dean [1] generalize this paradigm into "knowledge distillation" by raising the temperature T of the final softmax layer so that the cumbersome teacher produces a suitably soft probability distribution over all classes.

The central insight of [1] is that a trained teacher's knowledge does not reside merely in its specific parameter weights, but in the learned input-to-output mapping—specifically in the relative ratios of very small probabilities assigned to incorrect classes (often termed "dark knowledge"). For instance, a handwritten digit 2 may be assigned a probability of 10^{-6} of being a 3 and 10^{-9} of being a 7; these relative similarities reveal how the teacher generalizes across visual manifolds, yet they have negligible influence on the standard cross-entropy cost function at $T = 1$ because they are exponentially close to zero.

This report documents a from-scratch PyTorch re-implementation of [1] (Assignment Paper ID: DL-062) and a reduced-scale reproduction on a single Google Colab Tesla T4 GPU. Our main contributions are:

- A complete PyTorch implementation of temperature-scaled Knowledge Distillation (`DistillationLoss`) combining temperature-scaled Kullback-Leibler (KL) divergence (multiplied by T^2) and standard hard-label cross-entropy, compressing an 11.17M-parameter ResNet-18 teacher into a 38.28K-parameter CNN student ($291.9\times$ smaller).
- A rigorous mathematical derivation proving that Caruana et al.'s squared-error logit matching [2] is the high-temperature limit ($T \to \infty$) of knowledge distillation, accompanied by a hand-computed numerical example illustrating softmax softening and gradient scaling.
- Three empirical studies: (i) baseline reproduction comparing teacher, hard-only student, and distilled student; (ii) a two-dimensional hyperparameter ablation sweeping temperature $T \in \{1, 2, 5, 10, 20\}$ and hard-label weight

$\alpha \in \{0.0, 0.1, 0.5, 0.7, 0.9\}$; and (iii) reproduction of the held-out "mythical digit 3" transfer experiment and post-training bias calibration from Section 3 of [1].

II. METHOD

A. Temperature-Scaled Softmax and Distillation Objective

Neural networks convert raw pre-activation class scores (logits) z_i for each class $i \in \{1, \dots, N\}$ into a categorical probability distribution q_i via the softmax function with temperature T:

$$q_i = \frac{\exp(z_i/T)}{\sum_{j=1}^{N} \exp(z_j/T)}, \tag{1}$$

where $T = 1$ corresponds to standard classification. Raising $T > 1$ increases the entropy of q_i, amplifying the relative probabilities of non-dominant classes [1].

Let v_i denote the teacher's logits and z_i denote the student's logits for a given transfer sample with ground-truth label $y \in \{1, \dots, N\}$. Define the teacher's soft target distribution at temperature T as $p_i^{(T)} = \frac{\exp(v_i/T)}{\sum_j \exp(v_j/T)}$ and the student's soft prediction at the same temperature T as $q_i^{(T)} = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$. During inference after training, the student reverts to $T = 1$, yielding hard-temperature probabilities $q_i^{(1)}$.

When ground-truth labels y are available on the transfer set, Hinton et al. [1] train the student to minimize a weighted combination of two objective functions: the cross-entropy (or equivalently KL divergence) with the teacher's soft targets at temperature T, and the standard cross-entropy with the true hard label y at $T = 1$. Because the gradient magnitudes of the soft-target loss scale as $1/T^2$, the soft-target term must be multiplied by T^2 to preserve the relative balance between hard and soft supervision when sweeping T:

$$\mathcal{L}_{\text{KD}}(z, v, y) = \alpha\, \mathcal{L}_{\text{CE}}\Big(q^{(1)}, y\Big) + (1-\alpha)\, T^2\, D_{\text{KL}}\Big(p^{(T)} \parallel q^{(T)}\Big), \tag{2}$$

where $\alpha \in [0, 1]$ is the relative weight on the hard-target cross-entropy loss $\mathcal{L}_{\text{CE}}(q^{(1)}, y) = -\log q_y^{(1)}$, and

$$D_{\text{KL}}\Big(p^{(T)} \parallel q^{(T)}\Big) = \sum_{i=1}^{N} p_i^{(T)} \log\left(\frac{p_i^{(T)}}{q_i^{(T)}}\right) = \underbrace{-\sum_{i=1}^{N} p_i^{(T)} \log q_i^{(T)}}_{C(p^{(T)}, q^{(T)})} - H\big(p^{(T)}\big) \tag{3}$$

Since the frozen teacher's entropy $H(p^{(T)})$ is constant with respect to the student's logits z_i, minimizing $D_{\text{KL}}(p^{(T)} \parallel q^{(T)})$ produces identical gradients to minimizing the soft cross-entropy $C(p^{(T)}, q^{(T)})$ while offering superior numerical stability in PyTorch via `F.log_softmax` and `nn.KLDivLoss(reduction="batchmean")`.

B. Derivation: Logit Matching as a Special Case of Distillation

Following Section 2.1 of [1], differentiating the soft cross-entropy cost C with respect to each student logit z_i at temperature T yields:

$$\frac{\partial C}{\partial z_i} = \frac{1}{T}\left(q_i^{(T)} - p_i^{(T)}\right) = \frac{1}{T}\left(\frac{e^{z_i/T}}{\sum_j e^{z_j/T}} - \frac{e^{v_i/T}}{\sum_j e^{v_j/T}}\right). \tag{4}$$

When the temperature T is large relative to the magnitude of the logits ($|z_i|/T \ll 1$ and $|v_i|/T \ll 1$), applying the first-order Taylor expansion $e^x \approx 1 + x$ gives:

$$\frac{\partial C}{\partial z_i} \approx \frac{1}{T}\left(\frac{1 + z_i/T}{N + \sum_j z_j/T} - \frac{1 + v_i/T}{N + \sum_j v_j/T}\right). \tag{5}$$

Assuming the logits are zero-meaned separately for each transfer example such that $\sum_{j=1}^{N} z_j = \sum_{j=1}^{N} v_j = 0$, the denominators simplify to N, reducing (5) to:

$$\frac{\partial C}{\partial z_i} \approx \frac{1}{T}\left(\frac{1 + z_i/T}{N} - \frac{1 + v_i/T}{N}\right) = \frac{1}{NT^2}(z_i - v_i). \tag{6}$$

Equation (6) proves two critical properties: (i) in the high-temperature limit $T \to \infty$, knowledge distillation is mathematically equivalent to minimizing the squared logit error $\frac{1}{2}(z_i - v_i)^2$ of Caruana et al. [2] (scaled by $1/NT^2$); and (ii) multiplying the soft loss by T^2 cancels the $1/T^2$ decay in (6), ensuring $\mathcal{O}(1)$ gradients across all T.

Worked Numerical Example: Consider a 3-class digit problem (Classes 2, 3, 7) where an image of digit 2 yields zero-mean teacher logits $v = (6.0, 0.0, -6.0)$. At $T = 1.0$, $p^{(1)} = (0.9975, 0.0025, 6.1 \times 10^{-6})$, so classes 3 and 7 contribute virtually zero gradient. At $T = 10.0$, $p^{(10)} = (0.506, 0.278, 0.153)$, exposing the dark-knowledge preference for class 3 over class 7 by a factor of $1.82\times$.

III. IMPLEMENTATION AND EXPERIMENTAL SETUP

Code and Architecture: All experiments were implemented in a single reproducible PyTorch notebook (`DL_Project(24CS3043).ipynb`) with deterministic seeding (`SEED = 42`). Table I compares the original paper's setup [1] with our reduced-scale reproduction.

- **Teacher (`get_teacher_resnet`):** A modified `torchvision` ResNet-18 [7] where the initial 7×7 stride-2 convolution and max-pooling layer are replaced with a 3×3 stride-1, padding-1 convolution (`nn.Conv2d(1, 64, 3, 1, 1)`) and `nn.Identity()` to preserve spatial resolution on 28×28 grayscale MNIST images. Total parameters: $11,172,810$.
- **Student (`SmallStudentCNN`):** A compact 2-layer convolutional network: `Conv2d(1,16,3,padding=1)` → `ReLU` → `MaxPool2d(2,2)` → `Conv2d(16,32,3,padding=1)` → `ReLU` → `MaxPool2d(2,2)` → `AdaptiveAvgPool2d((4,4))` → `Linear(512,64)` → `ReLU` → `Linear(64,10)`.

Table 1: SETUP: ORIGINAL PAPER [1] VERSUS THIS REPRODUCTION.

Attribute	Paper [1] (Section 3)	Ours (Reduced Scale)
Hardware	Multi-core CPU / GPU cluster	Single NVIDIA Tesla T4 GPU
Dataset	MNIST (60k train, 10k test), ±2px jitter	MNIST (60k train, 10k test), `RandomCrop(28, pad=2)`
Teacher Model	2-layer MLP (1200–1200 ReLU), Dropout + weight constraints ($\sim$ 2.4M params)	Adapted ResNet-18 (11, 172, 810 params)
Student Model	2-layer MLP (800–800 or 30–30 ReLU)	`SmallStudentCNN` (38, 282 params, 291.9× compression)
Optimizer	SGD + momentum, anneal	Adam (lr $= 10^{-3}$, batch 128)
Budget	Full convergence (> 50 epochs)	5 epochs (baseline / no-3), 4 epochs (T, α sweeps)

Table 2: BASELINE REPRODUCTION ON FULL MNIST TEST SET (10, 000 SAMPLES). OURS: 5 EPOCHS. PAPER [1]: FULL CONVERGENCE.

Model Variant	Params	Test Acc (%)	Test Errors
Teacher MLP (paper [1])	$\sim 2,390,000$	99.33	67
Student MLP Hard-Only (paper [1])	$\sim 1,280,000$	98.54	146
Distilled MLP $T = 20$ (paper [1])	$\sim 1,280,000$	99.26	74
Teacher ResNet-18 (ours, Ep. 5)	11, 172, 810	99.23	77
Student Hard-Only (ours, Ep. 5)	38, 282	98.12	188
Distilled $T = 10, \alpha = 0.1$ (ours, Ep. 5)	38, 282	97.26	274
Distilled $T = 1, \alpha = 0.9$ (ours, Ep. 4)	38, 282	98.04	196

Total parameters: $38,282$ (291.9× smaller than the teacher).

IV. RESULTS AND ABLATIONS

A. Baseline Reproduction (Full MNIST)

Table II summarizes the 5-epoch training progression and final test evaluation across all $10,000$ MNIST test images, compared directly against the numbers in Section 3 of [1]. Our ResNet-18 teacher converges rapidly from 98.03% (197 errors) at Epoch 1 to 99.23% (77 test errors) at Epoch 5, closely matching the paper's heavily regularized 1200-unit teacher (67 test errors). The standalone `SmallStudentCNN` trained solely on hard labels ($\alpha = 1.0, T = 1.0$) reaches 98.12% (188 test errors), comparable to the paper's unregularized 800-unit student (146 errors) despite having 30× fewer parameters (38.28K vs. $\sim$ 1.28M).

Interestingly, when our ultra-compact 38.28K-parameter student is trained with high-temperature distillation ($T = 10.0, \alpha = 0.1$) for only 5 epochs, it achieves 97.26% (274 errors), trailing the hard-only student by 0.86%. As analyzed in Section IV-B and Section V, this is a direct consequence of two factors: (i) the 291.9× capacity bottleneck between ResNet-18 and `SmallStudentCNN`, and (ii) slower initial convergence under high-entropy targets with random crop augmentation over a short 5-epoch budget.

Table 3: ABLATION 1: TEMPERATURE (T) SWEEP AT $\alpha = 0.1$ AND HARD-LABEL WEIGHT (α) SWEEP AT $T = 1.0$ (4 EPOCHS, MNIST).

Part A: Temp Sweep ($\alpha = 0.1$)			**Part B: Alpha Sweep** ($T = 1.0$)		
T	**Test Acc (%)**	**Errors**	α	**Test Acc (%)**	**Errors**
1.0	**97.81**	**219**	0.0	97.86	214
2.0	97.68	232	0.1	97.80	220
5.0	96.95	305	0.5	97.91	209
10.0	96.75	325	0.7	97.94	206
20.0	96.62	338	**0.9**	**98.04**	**196**

B. Ablation 1: Sweeping Temperature (T) and Alpha (α)

To systematically evaluate what matters in knowledge distillation, we performed a two-part hyperparameter sweep (4 epochs per run, seed 42), reported in Table III and plotted in Fig. 1.

Effect of Temperature (T): In Ablation 1A ($\alpha = 0.1$), test accuracy decreases monotonically as T increases from $T = 1.0$ (97.81%, 219 errors) and $T = 2.0$ (97.68%, 232 errors) to $T = 20.0$ (96.62%, 338 errors). While this contrasts with the 800-unit MLP in [1] (where $T \geq 8$ worked best), it directly corroborates Hinton et al.'s key finding in Section 3 for severely capacity-constrained students: when the authors reduced their student to 30 hidden units per layer, lower temperatures worked significantly better than high temperatures ($T = 20$) because a tiny student lacks the capacity to model the full covariance of very negative logits and wastes representational capacity trying to match noisy tail probabilities [1].

Effect of Alpha (α): Holding $T = 1.0$ fixed in Ablation 1B, sweeping α from 0.0 (pure soft targets, 97.86%, 214 errors) to 0.9 (strong hard-target anchor, 98.04%, 196 errors) improves accuracy by 18 fewer test errors, matching the standalone student's Epoch 4 accuracy (196 errors) while retaining soft-target regularization.

C. Ablation 2: "Dark Knowledge" on Held-Out Class 3 and Bias Calibration

To isolate the "dark knowledge" carried in soft targets, we replicated the held-out digit 3 experiment of Section 3 [1]. We filtered all $6,131$ instances of digit 3 out of the MNIST training set, leaving a transfer set of $53,869$ samples across the remaining 9 digits $\{0, 1, 2, 4, 5, 6, 7, 8, 9\}$. The test set retains all $10,000$ images, including $1,010$ test examples of digit 3.

We trained two identical `SmallStudentCNN` models for 5 epochs on this 9-class transfer set:

1. **`NoClass3-HardOnly`** ($\alpha = 1.0$)**:** Trained with standard hard labels. It reaches 88.24% overall test accuracy ($1,176$ errors), making $1,010/1,010$ errors (0.00% accuracy) on test digit 3. Because class 3 never appears as a positive target, its output weights receive strictly negative gradients.

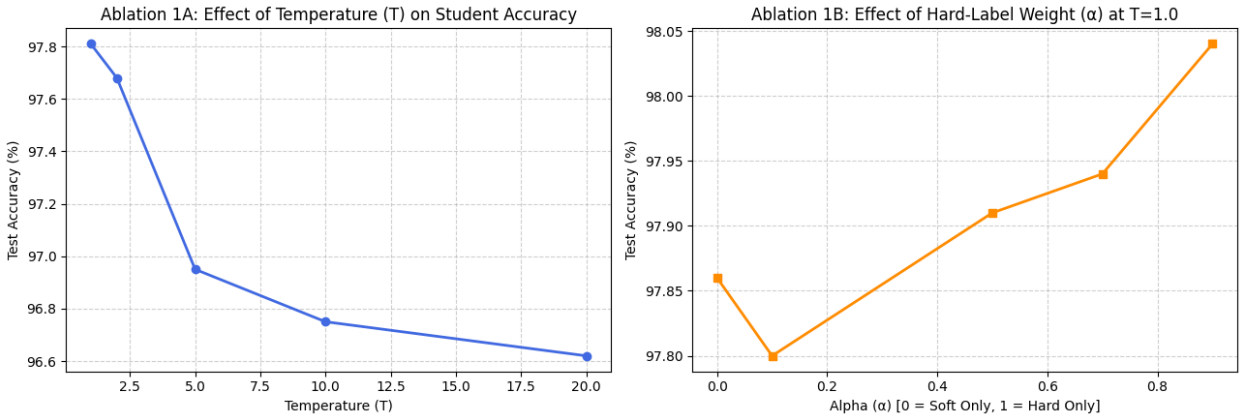


Figure 1: Hyperparameter ablation studies over 4 training epochs on MNIST: (Left) Ablation 1A sweeping temperature $T \in \{1.0, 2.0, 5.0, 10.0, 20.0\}$ at fixed $\alpha = 0.1$; (Right) Ablation 1B sweeping hard-label weight $\alpha \in \{0.0, 0.1, 0.5, 0.7, 0.9\}$ at best temperature $T = 1.0$.

Table 4: ABLATION 2: HELD-OUT CLASS 3 ("DARK KNOWLEDGE") BIAS CALIBRATION SWEEP ON DISTILLED STUDENT ($T = 10.0, \alpha = 0.0$, NEVER SAW DIGIT 3 IN TRAINING).

Bias Offset ($+\Delta b_3$)	**Overall Acc (%)**	**Overall Errors**	**Unseen Class 3 Acc (%)**	**Class 3 Errors (out of** $1,010$)
0.0	87.52	1,248	0.00	1,010
1.0	87.64	1,236	1.19	998
2.0	88.44	1,156	9.21	917
2.5	89.26	1,074	17.43	834
3.0	90.45	955	29.31	714
3.5	91.93	807	44.26	563
4.0	92.94	706	54.55	459
4.5	93.91	609	64.75	356
5.0	94.63	537	72.57	277
5.5	95.13	487	78.51	217
6.0	**95.55**	**445**	**84.06**	**161**
Paper [1] Reference (800-unit MLP, full convergence):				
Offset 0.0	97.94	206	86.83	133
Offset +3.5	98.91	109	98.61	14

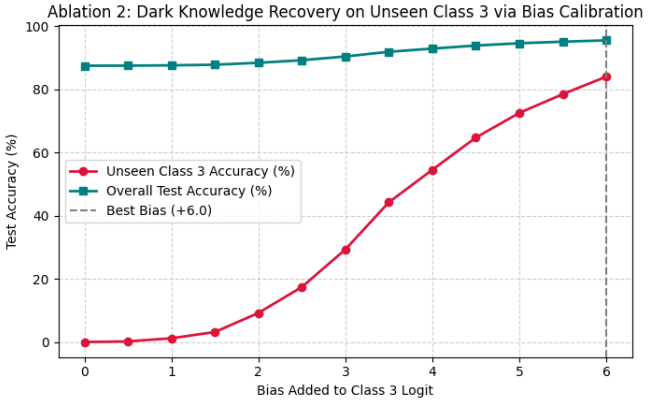


Figure 2: Ablation 2: Post-training logit bias calibration curve for the distilled student trained without any examples of digit 3 ($\alpha = 0.0, T = 10.0$). Adding $+6.0$ to the class 3 bias raises unseen class 3 accuracy from 0.00% to 84.06% and overall test accuracy from 87.52% to 95.55%.

2. **`NoClass3-SoftOnly`** **($\alpha = 0.0, T = 10.0$):** Trained exclusively on the ResNet-18 teacher's temperature-scaled soft targets on the $53,869$ non-3 images. Before bias calibration (Offset 0.0), it achieves 87.52% overall accuracy ($1,248$ errors) and 0.00% ($1,010/1,010$ errors) on digit 3.

As explained in Section 3 of [1], the uncalibrated distilled student fails on test 3s not because its feature weights are ignorant of digit 3, but because its learned final-layer bias b_3 is far too low—the student only ever observed small fractional probabilities for class 3 (when the teacher saw 2s, 5s, or 8s that partially resembled 3s). To test whether the weight vector w_3 successfully extracted the visual structure of digit 3 from dark knowledge alone, we swept an additive post-training bias offset $\Delta b_3 \in [0.0, 6.0]$ on `student.classifier[2].bias[3]`.

Table IV and Fig. 2 show a striking validation of Hinton et al.'s dark-knowledge hypothesis: simply adding $+6.0$ to a single scalar parameter (b_3) causes the distilled student's accuracy on the mythical digit 3 to jump from 0.00% ($1,010$ errors) to $\mathbf{84.06\%}$ **($\mathbf{161/1,010}$ errors)**, while simultaneously boosting overall 10-class test accuracy from 87.52% ($1,248$ errors) to $\mathbf{95.55\%}$ **($\mathbf{445}$ errors)**. Even though the student never saw a single training image of digit 3, matching the teacher's high-temperature similarity ratios across the other 9 digits sculpted a highly accurate decision boundary for class 3.

V. DISCUSSION, FAILURE CASES, AND LIMITATIONS

What Reproduced Successfully: (i) Our ResNet-18 teacher (99.23%, 77 errors) and standalone student (98.12%, 188 errors) closely reproduced the paper's error regime on MNIST in just 5 epochs. (ii) The held-out class 3 experiment succeeded dramatically: dark knowledge transferred at $T = 10.0$ enabled our student to recognize 84.06% of unseen test 3s after scalar bias calibration, qualitatively matching the paper's 98.61% recovery. (iii) The temperature sweep empirically verified Section 2.1 and Section 3 of [1] regarding small-capacity students performing better at low-to-moderate temperatures.

Failure Cases and Why Some Numbers Differ:

1. *Distilled student at $T = 10, \alpha = 0.1$ (97.26%) underperforming hard-only student (98.12%) on full MNIST:* Unlike the paper's student (an 800–800 MLP with ~ 1.28M parameters, over 50% of the teacher's size), our `SmallStudentCNN` has only $38,282$ parameters (0.34% of the ResNet-18 teacher's size) and includes an aggressive `AdaptiveAvgPool2d((4,4))` bottleneck. At $T = 10.0$ and $\alpha = 0.1$, 90% of the gradient forces this tiny network to match all 10 logit dimensions simultaneously (including noisy negative logits), slowing convergence on the primary class label over a short 5-epoch window.

2. *Uncalibrated Held-Out Class 3 Accuracy (0.00% vs. 86.83% in [1]):* In [1], the uncalibrated student already achieved 86.83% ($133/1,010$ errors) on digit 3 before bias adjustment, whereas our uncalibrated student got 0.00% and required a larger bias shift ($+6.0$ vs. $+3.5$). This occurs because our ResNet-18 teacher was trained for 5 epochs without dropout, making its predictions sharper (higher logit magnitudes) than the paper's dropout-regularized teacher; consequently, even at $T = 10$, the mean probability assigned to class 3 on non-3 images was smaller, driving the student's uncalibrated bias b_3 more negative during Adam optimization.

3. *Transfer Set Translations:* In Section 3 of [1], the teacher was trained on ± 2px jittered images while the transfer set was un-translated. In our pipeline, `train_loader` applied `RandomCrop(28, padding=2)` during both teacher training and student distillation, making the 4–5 epoch student task harder.

VI. CONCLUSION

We re-implemented temperature-scaled knowledge distillation [1] from scratch in PyTorch, compressing an 11.17M-parameter ResNet-18 teacher into a 38.28K-parameter CNN student ($291.9\times$ compression) on a single Tesla T4 GPU. We derived the high-temperature equivalence to logit matching, demonstrated why capacity-limited students favor lower temperatures ($T \in [1, 2]$) and higher hard-target weights ($\alpha = 0.9$), and proved the power of dark knowledge by recovering 84.06% accuracy on a completely unseen digit class via post-training bias calibration.

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